

	Healthcare Patient Risk Analysis	ASSIGNMENT - 1
		AI/ML INTERN

Dataset: <https://www.kaggle.com/datasets/prasad22/healthcare-dataset>

Dataset Provided Columns:

- Name
- Age
- Gender
- Blood Type
- Medical Condition
- Date of Admission
- Doctor
- Hospital
- Insurance Provider
- Billing Amount
- Room Number
- Admission Type
- Discharge Date
- Medication
- Test Results

Task 1 — EDA

1. Analyze distributions of **Age**, **Billing Amount**, and **Room Number**.
2. Visualize the frequency of **Medical Condition**, **Admission Type**, and **Medication**.

Task 2 — Supervised Learning

1. Prepare the dataset for predicting **Test Results** using the available features.
2. Split the dataset into training and testing sets.
3. Train a prediction model and evaluate performance using appropriate metrics.
4. Display predicted vs actual values for test data.

Task 3 — Unsupervised Learning

Anomaly Detection in Billing Amounts

1. Identify unusually high or low **Billing Amount** values compared to typical cases.
2. Detect entries that deviate significantly from expected billing patterns.
3. Mark the detected anomalies for review.

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4. Provide a short interpretation of anomalies (e.g., rare expensive cases).

Task 4 — AI Task (LLM-Based)

AI Doctor Recommendation Generator

1. Pass the patient's predicted test result and key attributes (Age, Medical Condition, Medication).
2. Generate a short AI-written doctor-style recommendation.
3. Include health advice based on the predicted result.
4. Produce one sample output.